



# **Master Presenter's Manual**

*for*

## **Network Administration**

**Curriculum Code: OV-1523**

**Release Date: June, 2024**  
**Version 1.0**

Amendment Record

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## 1. Introduction

This Presenter's Manual is to be used for the **Network Administration** course.

After the completion of this course, students will be able to:

- ❖ Explain networking and its uses
- ❖ Explain concepts of a networking model
- ❖ Learn the features of Cisco Internetwork Operating System (IOS)
- ❖ Define the importance of Loops in Switching
- ❖ Learn the importance of First Hop Redundancy
- ❖ Define the key terminologies used in network security
- ❖ Explain the purpose of Layer 3 Switches
- ❖ Define VPN and explain its types
- ❖ Learn the key components and elements involved in wireless networks
- ❖ Define Software Defined Access (SDA) and its role in network automation

## 2. General Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on the module along with a break-up of the hours being spent on the module.

This course is based on theory and practice lab session. A session is of two hours duration.

### Module Wise Session Allocation

| Module                                           | Online Hours | Skill Code |
|--------------------------------------------------|--------------|------------|
| Implementing and Administering Network Solutions | 40           | 1305       |

**Module Deliverables available on OnlineVarsity**

To aid the learning process, following are the deliverables

**Student Deliverables:**

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

| Icons                                                                               | Feature - Description/Functionality                                                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Download Book</b> - Student has the option to download the subject related e-book and read offline.                                                    |
|    | <b>Glossary</b> - Student can access a list of subject related specialized words with their definitions.                                                  |
|    | <b>FAQ</b> - Student can access frequently asked questions and their answers.                                                                             |
|    | <b>Practice 4 Me</b> - Student can test and evaluate their understanding of module related topics.                                                        |
|   | <b>Work Assignments</b> - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.                |
|  | <b>References</b> - Student can access additional subject related material for reading.                                                                   |
|  | <b>Feedback</b> - Student can provide feedback on the course material.                                                                                    |
|  | <b>Ask to Learn</b> - Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head. |

The table below shows the details of the module, the number of hours being spent on the module along with a break-up of the hours being spent on each session.

**3. Week Wise Schedule**

| Topic \ Week                                     |           | 1   | 2   | 3    | 4    | 5    |
|--------------------------------------------------|-----------|-----|-----|------|------|------|
| Implementing and Administering Network Solutions | <b>D1</b> | TL1 | TL5 | TL9  | TL13 | TL17 |
|                                                  | <b>D2</b> | TL2 | TL6 | TL10 | TL14 | TL18 |
|                                                  | <b>D3</b> | TL3 | TL7 | TL11 | TL15 | TL19 |
|                                                  | <b>D4</b> | TL4 | TL8 | TL12 | TL16 | TL20 |

Where,

**D** = Day, **TL** = Online Session

#### 4. Session Coverage

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Deliverables' Mapping                                                                                                                               |
|-------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | CCNA-TL1      | <p>All the topics as listed below from Session 1 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 1: Fundamentals of Networking</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain networking and its uses</li> <li>➤ Define and describe different criteria used to build a good network</li> <li>➤ Identify various devices used for networking</li> </ul>                                                                                                                                                         | <p><b><u>Implementing and Administering Network Solutions</u></b></p> <p>SG-Session 1<br/>XP-Session 1<br/>TG-Session 1</p>                         |
| 2           | CCNA-TL2      | <p>All the topics as listed below from Session 2 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 2: Models in Computer Networking</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain concepts of a networking model</li> <li>➤ Describe TCP/IP and OSI Model</li> <li>➤ Identify and explain the difference between TCP/IP and OSI Model</li> <li>➤ Explain various types of area networks</li> </ul>                                                                                                               | <p><b><u>Implementing and Administering Network Solutions</u></b></p> <p>SG-Session 2<br/>XP-Session 2<br/>TG-Session 2</p>                         |
| 3           | CCNA-TL3      | <p>All the topics as listed below from Session 3 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 3: Types of Addresses in Networking</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain the purpose and characteristics of MAC addresses</li> <li>➤ Define IP address and explain its significance in identifying devices on a network</li> <li>➤ Define and describe subnetting in IP addresses</li> <li>➤ Explain the concept of Classless Inter-Domain Routing (CIDR) and its impact on IP addressing</li> </ul> | <p><b><u>Implementing and Administering Network Solutions</u></b></p> <p>SG-Session 3<br/>XP-Session 3<br/>TG-Session 3</p>                         |
| 4           | CCNA-TL4      | <p>The <b>Try It Yourself</b> questions of Session 1 to Session 3 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p>                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><b><u>Implementing and Administering Network Solutions</u></b></p> <p>Session 1 to 3</p>                                                         |
| 5           | CCNA-TL5      | <p>All the topics as listed below from Session 4 and Session 5 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 4: Hardware and Software of CISCO Devices</u></b></p> <ul style="list-style-type: none"> <li>➤ Define the hardware and software components of routers and switches</li> <li>➤ Explain the Cisco Internetwork Operating System (IOS) and its features</li> </ul>                                                                                                                                                | <p><b><u>Implementing and Administering Network Solutions</u></b></p> <p>SG-Session 4 &amp; 5<br/>XP-Session 4 &amp; 5<br/>TG-Session 4 &amp; 5</p> |

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Deliverables' Mapping                                                                                                          |
|-------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Illustrate the steps to configure the clock, hostname and interfaces on Cisco devices</li> </ul> <b><u>Session 5: Introduction to Switching</u></b> <ul style="list-style-type: none"> <li>➤ Explain the role of Layer 2 switches in network communication</li> <li>➤ Define and describe the function of Layer 2 switches in forwarding frames within a local network</li> <li>➤ Define VLANs and illustrate the configuration of VLANs on Cisco switches</li> <li>➤ Explain the role of DTP and VTP in Cisco networks</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                |
| 6           | CCNA-TL6      | <p>The <b>Try It Yourself</b> questions from Session 4 and Session 5 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b><u>Implementing and Administering Network Solutions</u></b><br><br>Session 4 & 5                                            |
| 7           | CCNA-TL7      | <p>All the topics as listed below from Session 6 and Session 7 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <b><u>Session 6: Loop Avoidance in Cisco Switches</u></b> <ul style="list-style-type: none"> <li>➤ Explain the concept and importance of loops in switches and the Spanning Tree Protocol (STP)</li> <li>➤ Explain the purpose and functionality of BPDU Guard</li> <li>➤ Explain EtherChannel and its advantages</li> <li>➤ Elaborate on Port Aggregation Protocol (PAGP) and Link Aggregation Control Protocol (LACP)</li> </ul> <b><u>Session 7: Routing Technologies</u></b> <ul style="list-style-type: none"> <li>➤ Define routing and explain its fundamentals</li> <li>➤ Describe administrative distance, autonomous systems and longest prefix match</li> <li>➤ Illustrate the configuration of static routing using IPv4 and IPv6 addresses</li> <li>➤ Illustrate the configuration of floating static routing using IPv4 and IPv6 addresses</li> <li>➤ Explain the concepts of single area OSPF and multi-area OSPF</li> <li>➤ Illustrate the configuration of single area OSPF and multi-area OSPF</li> </ul> | <b><u>Implementing and Administering Network Solutions</u></b><br><br>SG-Session 6 & 7<br>XP-Session 6 & 7<br>TG-Session 6 & 7 |
| 8           | CCNA-TL8      | <p>All the topics as listed below from Session 8 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <b><u>Session 8: First Hop Redundancy</u></b> <ul style="list-style-type: none"> <li>➤ Summarise the concept of First Hop Redundancy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b><u>Implementing and Administering Network Solutions</u></b><br><br>SG-Session 8<br>XP-Session 8<br>TG-Session 8             |

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Deliverables' Mapping                                                                                                 |
|-------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Explain the importance and requirement of First Hop Redundancy and its protocols</li> <li>➤ Define and describe HSRP, VRRP and GLBP</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                       |
| 9           | CCNA-TL9      | The <b>Try It Yourself</b> questions of Session 6 to Session 8 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <u><b>Implementing and Administering Network Solutions</b></u><br><br>Session 6-8                                     |
| 10          | CCNA-TL10     | <p>All the topics as listed below from Session 9 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 9: Protocols for Network Management</u></b></p> <ul style="list-style-type: none"> <li>➤ Define network management</li> <li>➤ Explain the purpose and functioning of Network Time Protocol (NTP)</li> <li>➤ Illustrate the configuration of Dynamic Host Configuration Protocol (DHCP) on Cisco routers</li> <li>➤ Summarise Domain Name System (DNS) and its role in network management</li> <li>➤ Explain the basics of the TFTP and its uses in network management</li> </ul>               | <u><b>Implementing and Administering Network Solutions</b></u><br><br>SG-Session 9<br>XP-Session 9<br>TG-Session 9    |
| 11          | CCNA-TL11     | <p>All the topics as listed below from Session 10 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 10: Network Security Fundamentals</u></b></p> <ul style="list-style-type: none"> <li>➤ Define the key terminologies used in network security</li> <li>➤ Explain the importance of implementing access control measures to protect network resources</li> <li>➤ Define digital certificates and explain their purpose in network security</li> <li>➤ Explain the syntax and structure of ACLs</li> <li>➤ Explain DHCP snooping and its role in preventing rogue DHCP server attacks</li> </ul> | <u><b>Implementing and Administering Network Solutions</b></u><br><br>SG-Session 10<br>XP-Session 10<br>TG-Session 10 |
| 12          | CCNA-TL12     | The <b>Try It Yourself</b> questions from Session 9 and Session 10 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <u><b>Implementing and Administering Network Solutions</b></u><br><br>Session 9 & 10                                  |
| 13          | CCNA-TL13     | <p>All the topics as listed below from Session 11 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.</p> <p><b><u>Session 11: IPv6</u></b></p> <ul style="list-style-type: none"> <li>➤ Define IPv6 and explain the reasons for implementing IPv6</li> <li>➤ Explain differences between IPv4 and IPv6 addressing schemes</li> </ul>                                                                                                                                                                                                                                                                                    | <u><b>Implementing and Administering Network Solutions</b></u><br><br>SG-Session 11<br>XP-Session 11<br>TG-Session 11 |



| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Deliverables' Mapping                                                                                                                |
|-------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Explain the Neighbor Discovery Protocol (NDP) used in IPv6</li> <li>➤ Define DHCPv6 and describe its role in IPv6 network configuration</li> <li>➤ Describe the principles of static routing with IPv6</li> <li>➤ Explain Open Shortest Path First (OSPF v3) and its implementation in IPv6 networks</li> </ul>                                                                                                                                                                              |                                                                                                                                      |
| 14          | CCNA-TL14     | The <b>Try It Yourself</b> questions from Session 11 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.                                                                                                                                                                                                                                                                                                                                                                               | <u><b>Implementing and Administering Network Solutions</b></u><br><br>Session 11                                                     |
| 15          | CCNA-TL15     | All the topics as listed below from Session 12 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.<br><br><u><b>Session 12: Layer Switches</b></u> <ul style="list-style-type: none"> <li>➤ Explain the concept and purpose of Layer 3 switches in a network environment</li> <li>➤ Explain CEF (a fast and efficient switching mechanism used by Cisco devices)</li> <li>➤ Explain the concept of routing between VLANs</li> <li>➤ Illustrate Switch VLAN Interfaces (SVI)</li> </ul> | <u><b>Implementing and Administering Network Solutions</b></u><br><br>SG-Session 12<br>XP-Session 12<br>TG-Session 12                |
| 16          | CCNA-TL16     | The <b>Try It Yourself</b> questions from Session 12 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.                                                                                                                                                                                                                                                                                                                                                                               | <u><b>Implementing and Administering Network Solutions</b></u><br><br>Session 12                                                     |
| 17          | CCNA-TL17     | All the topics as listed below from Session 13 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.<br><br><u><b>Session 13: Virtual Private Networks</b></u> <ul style="list-style-type: none"> <li>➤ Define VPNs and explain their purpose in network communication</li> <li>➤ Describe the aspects of VPN secure system</li> <li>➤ Explain the advantages of using a VPN</li> <li>➤ Describe various types of VPNs</li> </ul>                                                        | <u><b>Implementing and Administering Network Solutions</b></u><br><br>SG-Session 13<br>XP-Session 13<br>TG-Session 13                |
| 18          | CCNA-TL18     | The <b>Try It Yourself</b> questions from Session 13 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.                                                                                                                                                                                                                                                                                                                                                                               | <u><b>Implementing and Administering Network Solutions</b></u><br><br>Session 13                                                     |
| 19          | CCNA-TL19     | All the topics as listed below from Session 14 and Session 15 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.<br><br><u><b>Session 14: Wireless Technologies</b></u> <ul style="list-style-type: none"> <li>➤ Identify the key components and elements involved in wireless networks</li> </ul>                                                                                                                                                                                    | <u><b>Implementing and Administering Network Solutions</b></u><br><br>SG-Session 14 & 15<br>XP-Session 14 & 15<br>TG-Session 14 & 15 |

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Deliverables' Mapping                                                                            |
|-------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Define the roles and functions of Access Points (AP) and Wireless LAN Controllers (WLC)</li> <li>➤ Explain different wireless security protocols</li> <li>➤ Define WEP, WPA and WPA2 and explain its role in wireless security</li> <li>➤ Illustrate the configuration of a Wireless Local Area Network (WLAN)</li> </ul> <p><b><u>Session 15: Automation and Network Programmability</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain how automation improves network efficiency and reliability</li> <li>➤ Define Software Defined Access (SDA) and its role in network automation</li> <li>➤ Describe how REST-based APIs enable network automation</li> <li>➤ Explain Cisco Digital Network Architecture (DNA)</li> <li>➤ Describe the importance of effective configuration management</li> <li>➤ Explain the advantages of using JSON in network automation processes</li> </ul> |                                                                                                  |
| 20          | CCNA-TL20     | The <b>Try It Yourself</b> questions from Session 14 and Session 15 of <i>Implementing and Administering Network Solutions</i> book should be covered in this session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><b><u>Implementing and Administering Network Solutions</u></b></p> <p>Session 14 &amp; 15</p> |

## 5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

### ➤ Deliverables for Students:

|                           |                                                                         |
|---------------------------|-------------------------------------------------------------------------|
| <b>Learner's Guide</b>    | eBook consisting of the knowledge to be imparted (wherever applicable)  |
| <b>Assignments</b>        | Case study/scenario based problems to be solved by learners             |
| <b>Article</b>            | Additional knowledge in the form of articles that are referred from Web |
| <b>Practice Questions</b> | Additional practice questions for learners at the end of each session   |

| Module                                           | Book Code                 | Book Name                                                |
|--------------------------------------------------|---------------------------|----------------------------------------------------------|
| <b>KIT</b>                                       | <b>OV-1523-KIT01-01-A</b> | <b>Kit of 1 title for OV-1523 Network Administration</b> |
| Implementing and Administering Network Solutions | AINNIANSN10323E000        | Implementing and Administering Network Solutions         |

### ➤ Deliverables for Faculty

|                                |                                                                            |
|--------------------------------|----------------------------------------------------------------------------|
| <b>Presenter's Manual</b>      | A training delivery plan for the trainers                                  |
| <b>Trainer's Guide</b>         | A session wise guide for the trainers                                      |
| <b>Classroom Presentations</b> | Session wise PowerPoint presentations as teaching aid, wherever applicable |

| Module Name                                      | Book Code                 | Book Name                                                |
|--------------------------------------------------|---------------------------|----------------------------------------------------------|
| <b>KIT</b>                                       | <b>OV-1523-KIT01-01-A</b> | <b>Kit of 1 title for OV-1523 Network Administration</b> |
| Implementing and Administering Network Solutions | AINNIANSN10323E000        | Implementing and Administering Network Solutions         |

## 6. Library References

|                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ➤ <b>Implementing and Administering Cisco Solutions: 200-301 CCNA Exam Guide: Begin a successful career in networking with CCNA 200-301 certification</b> by Glen D. Singh |
| ➤ <b>CCNA 200-301 Official Cert Guide Library</b> by Wendell Odom                                                                                                          |

## 7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination (Objective)
- Practical Examination

### 7.1 Award of Grades

**Overall Weighted Marks > = 40% but < 60%** qualifies for **PASS**

**Overall Weighted Marks > = 60% but < 75%** qualifies for **CREDIT** and

**Overall Weighted Marks > = 75%** qualifies for **DISTINCTION**

**Note:** To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in each of the elements of the Examination; otherwise he/she will be declared as 'Referred'.

### 7.2 Final Examination 1 (E1)

Below is the list of modules to be covered in Final Examination along with their weightages:

#### Final Examination 1 (E1)

**Marks:** 50

**Duration:** 1 hour

**Type:** Objective

| S# | Name of the Subject                              | Marks |
|----|--------------------------------------------------|-------|
| 1. | Implementing and Administering Network Solutions | 50    |

#### Final Examination 2 (R1)

**Marks:** 50

**Duration:** 2 hours

**Type:** Practical

| S# | Name of the Subject                              | Marks |
|----|--------------------------------------------------|-------|
| 1. | Implementing and Administering Network Solutions | 50    |

### Evaluation Details

**OES-50%**

**Practical-50%**

## **8. Classroom and Lab Pre-requisites**

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The requirements for conducting the sessions have been described as follows:

### **8.1 Software Requirements**

Microsoft Windows 11

Graphical Network Simulator 3(GNS3) or any other Router Simulation Software

### **8.2 Recommended Hardware**

Intel Core i5/i7 Processor or higher

8 GB or 16 GB RAM

1 TB Hard Disk space

Mouse

Keyboard

### **8.3 Other Requirements (For Classroom)**

Computer for giving demonstrations wherever applicable

LCD projector

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